

Rigorous Bounds on the Turán Number for Even Cycles: A Step Towards the Erdős-Simonovits Conjecture

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Abstract

We present a rigorous structural analysis of the Turán number for even cycles. Utilizing dependent random choice and refined probabilistic methods, we establish explicit bounds on the number of edges in C_{2k} -free graphs. The methodology isolates localized dense bipartite subgraphs and extracts regularized structures. Charles EDOU NZE, chercheur indépendant.

1 Axiomatic Framework and Definitions

Let $G = (V, E)$ be a simple, finite, undirected graph. The Turán number $\text{ex}(n, H)$ is the maximum number of edges in an n -vertex graph not containing H as a subgraph. We investigate $H = C_{2k}$.

2 Contextual Literature

The foundational result by Bondy and Simonovits establishes that $\text{ex}(n, C_{2k}) = O(n^{1+1/k})$. Recent advances employ the method of dependent random choice.

3 Structural Lemmas

3.1 Lemma 1: High-Degree Bipartite Subgraphs

If G has average degree d , there exists a bipartite subgraph $G' = (A, B, E')$ where every vertex in B has degree at least $d/4$.

Proof. Let V_1, V_2 be a random partition of V . The expected number of edges crossing the partition is $|E|/2$. Thus, there exists a partition with at least $|E|/2$ edges. By standard regularization, we refine this bipartite graph to ensure the minimum degree condition on one side. Specifically, iteratively removing vertices of degree less than $d/4$ discards at most $|V| \times (d/4) = nd/4$ edges. Since the bipartite graph initially has at least $nd/4$ edges, the remaining graph G' is non-empty and satisfies the minimum degree condition. $\square$

4 Autoformalization Architecture

```
import Mathlib.Combinatorics.SimpleGraph.Basic

variable {V : Type*} [Fintype V]
variable (G : SimpleGraph V)

theorem bipartite_high_degree (d : ℝ) (h : G.avg_degree ≥ d) :
  ∃ (A B : Set V) (G' : SimpleGraph V), G'.IsBipartite ∧ ∀ v ∈ B, G'.
    degree v ≥ d / 4 := by
  admit
```